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Compact mode converters in thin-film lithium niobate integrated platforms

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Mode converters, crucial elements within photonic integrated circuits (PICs) designed for multimode optical transmission and switching systems, present a challenge due to their bulky structures in thin-film lithium niobate (TFLN) integrated platforms, which are incompatible with the compact and efficient nature desired for dense PICs. In this work, we propose TE_1-TE_0, TE_2-TE_0, and TE_3-TE_0 mode converters in shallowly etched TFLN, within small footprints. The experimental results show that the insertion loss is 0.4 dB, 0.6 dB, and 0.5 dB for the compact TE_1-TE_0, TE_2-TE_0, and TE_3-TE_0 mode converters, respectively, and these devices can be operated within a wide 1 dB bandwidth (BW) over 100 nm. This work facilitates the development of low-loss, broadband, and compact monolithically integrated photonic devices for future multimode communication networks in TFLN integrated platforms.

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Given the substantial expansion of an optical communication system scale to accommodate extensive data transmission, there is a significant demand for photonic devices characterized by compact dimensions and low loss [1,2]. Mode converters play a crucial role in modern photonic integrated circuits (PIC), by enabling efficient and reliable conversion of optical signals among different spatial modes [3]. They are essential for mitigating mode mismatches and enabling seamless interfacing between optical components operating with different spatial modes, for mode-division multiplexing and demultiplexing [4,5]. In addition to their role in high-capacity data transmission, mode converters also find applications in a high-order mode that involved a nonlinear interaction, e.g., modal phase-matched wavelength conversion [6,7]. Thus, compact and efficient mode converts are highly desired in PICs.

Thin-film lithium niobate (TFLN), a rapidly advancing integrated platform, has gained substantial attention for its potential to revolutionize optical communications [8,9]. Its strong electro-optic effect enables efficient modulation and switching of light, making it ideal for high-speed optical signal processing [10,11]. Meanwhile, it exhibits low optical losses within a wide transparent window spanning from the near-ultraviolet to mid-infrared wavelength range, enabling the realization of low-loss on-chip optical devices, compatible with a broad wavelength range. Up to now, a variety of high-performance functional devices have been demonstrated in thin-film lithium niobate, such as polarization rotators, polarization beam splitters, edge couplers, and mode converters [12,13]. However, due to the relatively lower refractive index of TFLN, as well as the fabrication difficulty, it is challenging to make these devices have high compactness.

Various types of mode converters have been investigated in mature integrated platforms, e.g., silicon [14,15]. One commonly employed type of mode converters is the asymmetric directional coupler, which gradually transforms one mode into another [16]. Asymmetric directional couplers typically exhibit low insertion loss, allowing for efficient mode conversion and maintaining a broad bandwidth (BW), but they often require longer device lengths, resulting in a large footprint [17]. This mode converting scheme has been applied by using silicon nitride on TFLN loaded waveguides or a polymer on TFLN BIC waveguides, but they either require an additional material layer or are restrictive to the waveguide dimension, extensively raising inflexibility and complexity [18,19]. A monolithic TFLN mode converter based on directional coupling has been reported to realize TE_0-TE_1 mode conversion, but it requires a very long coupling region, which is not desired for dense PICs [13]. Another common type of mode converters is multimode interference (MMI), which has also been explored in TFLN [20,21]. MMI couplers offer a rather compact footprint, but they often introduce higher insertion loss compared to the directional couplers and also suffer from narrower bandwidths due to the inherent multimode interference effects [22,23]. A high-performance multiplexer with an anisotropic TFLN waveguide has been reported to realize TE_1-TE_0,TE_2-TE_0, and TE_3-TE_0 mode conversion, but their proposed asymmetric corner-cutting structure makes the fabrication of the device very difficult [24]. Thus, it still remains a challenge to make high-performance compact mode converters in TFLN.

In this Letter, we employ assisted waveguides and slots for the purpose of beam shaping to achieve compact mode converters,

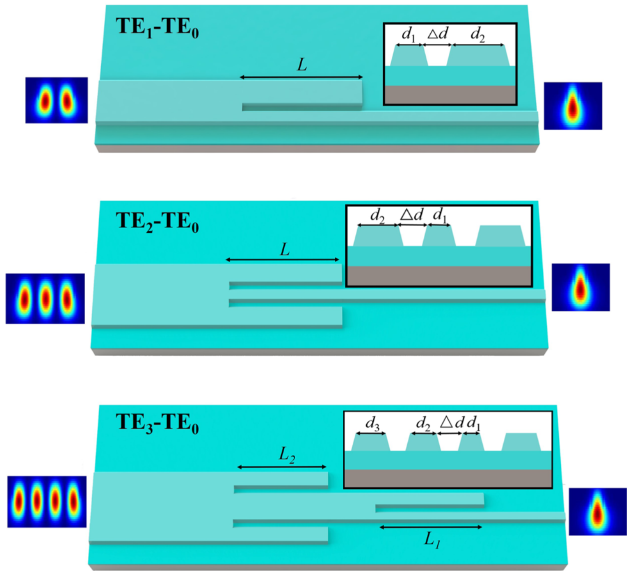

Fig. 1. Schematics of TE_1-TE_0, TE_2-TE_0, and TE_3-TE_0 mode converters with the corresponding mode profiles. The insets show the cross section of the mode converters.

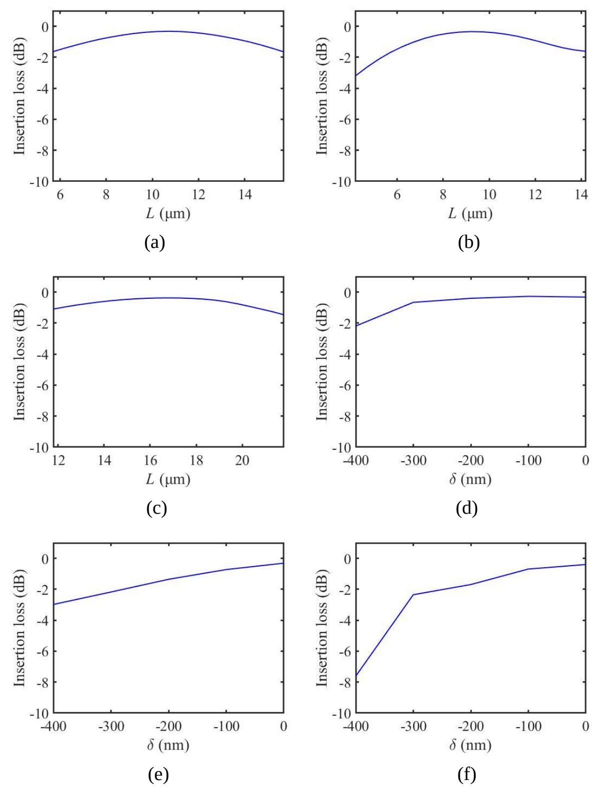

Fig. 2. Simulated insertion loss as a function of the coupling length in (a) TE_1-TE_0, (b) TE_2-TE_0, and (c) TE_3-TE_0 mode converters. The impact of manufacturing tolerances in (d) TE_1-TE_0, (e) TE_2-TE_0, and (f) TE_3-TE_0 mode converters.

which can be directly fabricated in z-cut TFLN integrated platforms, without additional material layers. The devices are bidirectional, enabling an efficient and broadband conversion between the TE high-order modes and the TE fundamental mode (TE_0), including the first-order (TE_1), second-order (TE_2), and third-order (TE_3) modes. The schematics of the TE_1-TE_0, TE_2-TE_0, and TE_3-TE_0 mode converters and the corresponding mode profiles are shown in Fig. 1. These devices are made of 300-nm-thick LN rib waveguides with 300-nm-thick LN slab on insulator, and the sidewall has an angle of 70°, according to the fabrication.

The narrow slot structures have been applied to various applications, such as compact polarization beam splitters, broadband wavelength conversion, and efficient mode conversion [25,26]. When employing slots with assisted waveguides for mode conversion, such structure can be considered as a power coupler and a phase shifter, simultaneously. For example, when converting from the TE_1 mode to the TE_0 mode, the origin TE_1 mode is split into two TE_0 modes with a phase difference of π. Subsequently, the slot structure is able to couple the TE_0 mode in the assisted waveguide to the output waveguide while changing the phase at the same time to make it identical. Conversely, when converting the mode from TE_0 to TE_1, the TE_0 mode is split into two TE_0 modes of equal power in the incident waveguide and the assisted waveguide while inducing a phase difference of π between the two beams synchronously during the cross-power coupling. Ultimately, the recombination of the two anti-phase TE_0 modes culminates in the generation of a TE_1 mode, facilitated by their similar mode profiles.

To design the TE_1-TE_0 mode converter, we fix the slot width Δd as 420 nm, considering the fabrication resolution. The width of the input single-mode waveguide and the assisted waveguide for beam shaping is set as $d_1 = 500$ nm and $d_2 = 800$ nm. The coupling length can be roughly predicted according to the coupled mode theory and the effective index method [26]. Then, we can use these parameters to build a model in Lumerical FDTD and optimize the coupling length. As can be seen in Fig. 2(a), the optimal coupling length is 10.3 μm, leading to a low insertion loss of 0.3 dB. The transmission efficiency is not very sensitive to the coupling length.

When converting the mode from the TE_2 mode to the TE_0 mode, we employ dual slots by two assisted waveguides, symmetrically placed beside the central waveguide. When converting the mode from TE_2 to TE_0, the TE_2 mode is split into three TE_0 modes, and the dual-slot structure is able to couple the TE_0 modes in the side waveguides to the central waveguide and vary the phase at the same time. Conversely, when converting the mode from TE_0 to TE_2, the TE_0 mode is split into three beams while inducing a phase difference of π between the split beams and the original beam, synchronously during the propagation. The recombination of the three TE_0 modes results in the production of a TE_2 mode.

To design the TE_2-TE_0 mode converter, the slot width and the width of the input single-mode waveguide and the assisted waveguide are selected the same as those in the TE_1-TE_0 mode converter. Figure 2(b) shows the optimization of the coupling length, and the minimum insertion loss is 0.3 dB at 9 μm.

To make a TE_3-TE_0 mode converter, we connect a TE_3-TE_1 mode converter with a TE_1-TE_0 mode converter. The TE_3-TE_1 mode converter is similar to the TE_2-TE_0 mode converter, where dual slots by assisted waveguides are employed symmetrically beside the central multimode waveguide. The TE_3 mode is split into two TE_0 modes and one TE_1 mode, and the TE_0 mode in the assisted waveguides couples to the central waveguide becoming the TE_1 mode. Then, the TE_1 is converted to the TE_0 mode through the slot structure. Conversely, when converting

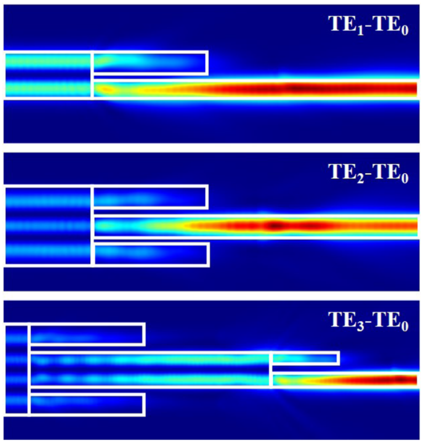

Fig. 3. Energy distribution in the TE_1-TE_0, TE_2-TE_0, and TE_3-TE_0 mode converters.

the mode from TE_0 to TE_3, the TE_1 mode is split into three beams, two of which in the assisted waveguide are the TE_0 modes. The recombination of the TE_1 modes with the two TE_0 modes results in the generation of a TE_3 mode.

To design the TE_3-TE_0 mode converter, we use the above optimized TE_1-TE_0 mode converter and further optimize the parameters of the TE_3-TE_1 mode converter. The slot width Δd is 420 nm, and the width of the two assisted waveguides for beam shaping is set as $d_3 = 0.8$ µm. Figure 2(c) shows the optimization of the coupling length of the TE_3-TE_1 mode converter, and the minimal insertion loss is 0.4 dB at 17.4 µm. Figures 2(d)–2(f) show the impact of manufacturing tolerances in the TE_1-TE_0, TE_2-TE_0, TE_3-TE_0 mode converters.δ represents the change in the waveguide width due to manufacturing tolerance. From the figure, the three devices still maintain relatively low insertion losses within a tolerance range of 100 nm.

Figure 3 shows the energy distribution in the three designed devices, indicating that these devices work effectively. Figure 4 plots the corresponding simulated transmission spectrum from 1400 to 1700 nm, showing that these devices can be operated within a wide wavelength range. In addition, the proposed scheme of the TE_1-TE_0 mode converter can be extended to achieve any adjacent odd-order and even-order mode conversion, and the proposed scheme of the TE_2-TE_0 mode converter can be extended to achieve a mode conversion with either adjacent even orders or adjacent odd orders. Thus, one can cascade these mode converters to achieve any higher-order mode to low-order mode transformation, and these devices are also bidirectional.

To verify the above simulation, we fabricate the three designed mode converters in a z-cut TFLN, with a 600-nm-thick $LiNbO_3$ layer on a 4.7-µm-thick buried oxide layer. The structures are patterned on hydrogen silsesquioxane (HSQ) through electron beam lithography and are transferred to TFLN through ion beam etching (IBE) to shallowly etched 300 nm thick $LiNbO_3$. Figure 5 shows the scanning electron microscopy (SEM) images of the fabricated mode converters. To accurately evaluate the device performances, every waveguide is composed of a pair of connected mode converters, which can transform the TE_0 mode to a high-order TE mode and transform back to the TE_0 mode for a robust single-mode fiber-to-chip coupling with the standard single-mode fiber-based measurement setup.

Figure 6(a) shows the schematic of the measurement setup. A tunable continuous-wave laser source (TLS) launches light

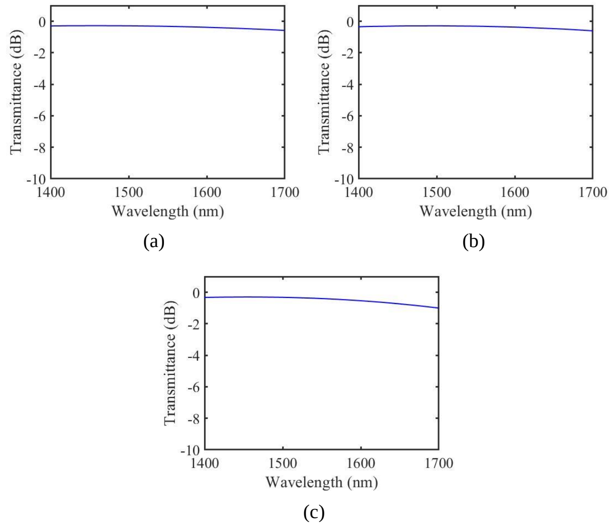

Fig. 4. Simulated transmittance of the (a) TE_1-TE_0, (b) TE_2-TE_0, and (c) TE_3-TE_0 mode converters.

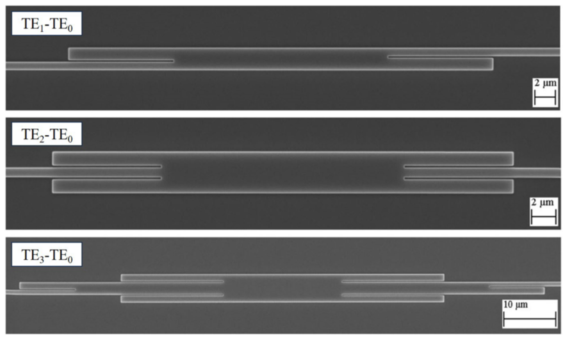

Fig. 5. SEM images of the fabricated TE_1-TE_0, TE_2-TE_0, and TE_3-TE_0 mode converters in TFLN.

into a polarization controller (PC), which makes the light TE polarized. The TE polarized light is then coupled in and out of the device through edge coupling with a pair of lensed fibers. The output signal is detected by a powermeter (PM), synchronized with the TLS to measure the transmission spectrum of the devices.

Figures 6(b)–6(d) show the normalized transmission spectrum of single TE_1-TE_0, TE_2-TE_0, and TE_3-TE_0 mode converters from 1500 to 1630 nm, respectively. From the smoothened curves, the minimum insertion loss of TE_1-TE_0, TE_2-TE_0, and TE_3-TE_0 mode converters is 0.4 dB, 0.6 dB, and 0.5 dB, respectively. The transmission spectrum of the TE_1-TE_0 mode converter looks very flat, exhibiting ultra-broad bandwidth over the whole spectrum. However, the conversion efficiency of the TE_2-TE_0 and TE_3-TE_0 mode converters is gradually increasing with the wavelength, and the spectrum profiles show redshifts, compared to the simulated results. This could be due to the linewidth variation, induced by the nanofabrication imperfection. Nevertheless, their transmission spectra still show a broad 1 dB bandwidth over a 100 nm wavelength range, indicating that the devices have high fabrication tolerance.

We compare our experimental results with the other reported mode converters in TFLN integrated platforms, in terms of insertion loss, 1 dB bandwidth, footprint, and the type of integration,

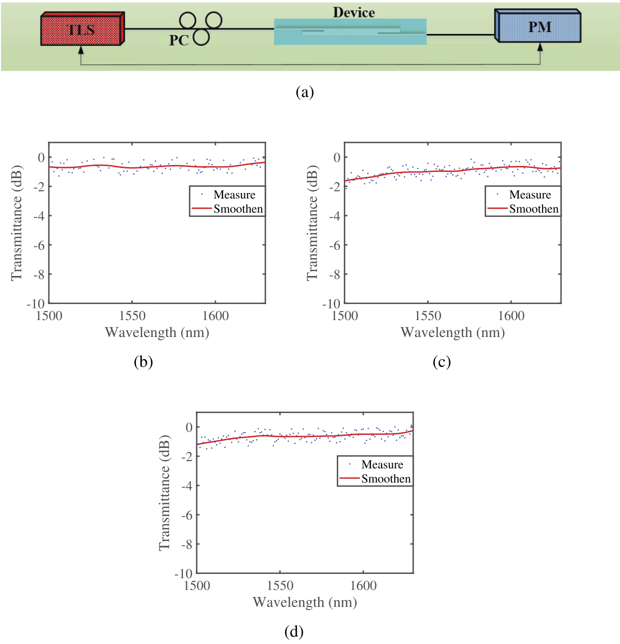

Fig. 6. (a) Schematic of the measurement setup. Normalized measured transmission spectrum for single (b) TE_1-TE_0, (c) TE_2-TE_0, and (d) TE_3-TE_0 mode converters. The blue points show the measured data, and the red line shows the smoothened curve.

Table 1. Comparison of the Reported TFLN Mode Converters with the Devices Demonstrated in This Work[a]

Conversion	Loss (dB)	BW (nm)	L (μm)	Type	Ref.
TE_0-TE_1	0.57	40	8	Loaded	[18]
TE_0-TE_2	0.73	40	16	Loaded	[18]
TE_0-TE_3	0.36	40	23	Loaded	[18]
TM_0-TM_1	1.4	70	125	BIC	[27]
TM_0-TM_2	1.35	70	157	BIC	[27]
TM_0-TM_3	1.35	70	143	BIC	[27]
TE_0-TE_1	0.4	100	550	Etched	[13]
TE_1-TE_0	0.2	80	100	Etched	[24]
TE_2-TE_0	0.2	80	100	Etched	[24]
TE_3-TE_0	0.2	80	100	Etched	[24]
TE_1-TE_0	0.4	130	10.3	Etched	TW
TE_2-TE_0	0.6	130	9.2	Etched	TW
TE_3-TE_0	0.5	130	10.3 + 17.4	Etched	TW

[a]In terms of insertion loss, 1 dB bandwidth (BW), length, and type of integration; TW, this work.

as shown in Table 1. Generally, the devices demonstrated in this work exhibit overall excellent performances, including low loss and broad bandwidth with high compactness, among all the reported ones. It is also worth mentioning that these slots and assisted waveguide-based mode converters can be monolithically integrated with other photonic devices directly fabricated in TFLN for high-dimensional optical communication, without any other assisted material layers.

In conclusion, we numerically simulate and experimentally demonstrate compact TE_0-TE_1, TE_0-TE_2, and TE_0-TE_3 mode converters with short lengths. These mode converters exhibit low insertion loss with a wide operational bandwidth. This work paves the way for high-dimensional mode-division multiplexing in TFLN, holding great promise for addressing the escalating bandwidth requirements in optical communication systems and achieving dense photonic integrated circuits.

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Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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